

Mathematical Foundations of Qubit Kinematics on the Bloch Sphere

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1. ABSTRACT & THEORETICAL FOUNDATION

We introduce a geometric parameterization of two-level quantum states onto the unit sphere S^2 , defining state vectors as linear combinations of orthogonal eigenstates with relative phase and polar angle.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"While a normal computer bit is like a light switch stuck at ON (1) or OFF (0), a qubit is like a sphere where the switch can point anywhere on its surface at once."

3. Mathematical Formulation & Unitary Rule

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

$\alpha, \beta \in \mathbb{C}$ are complex probability amplitudes satisfying the normalization constraint $|\alpha|^2 + |\beta|^2 = 1$. The angles θ and ϕ specify coordinates on the Bloch sphere.

4. Step-by-Step Circuit Sequence

Initialized in $|0\rangle$ ground state on physical transmon, ion-trap, or photonic hardware before microwave or laser drive pulses.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit
import numpy as np

# Create single qubit circuit
qc = QuantumCircuit(1)

# Rotate qubit to |+> state using Hadamard
qc.h(0)
```

6. Computational Complexity & Speedup

Classical Time:

$O(1)$ bit state

Quantum Time:

$O(1)$ state vector (continuous manifold)

Speedup Class:

Foundational

7. Key Applications & Future Research Directions

- Quantum memory registers
- Quantum sensing and atomic magnetometry
- Fundamental carrier for all quantum algorithms

Future Work:

Extension to higher-dimensional qudit state spaces and geometric phase holonomies for fault-tolerant logic gates.